



6319 - Measuring ionizing photon production efficiency and characterizing the bursty star formation history of dwarf galaxies at $0.5 < z < 2.4$

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST continues to amaze us by gazing at faint galaxies from the epoch of reionization (EoR). Nevertheless, with many hours of observations, the fundamental question that remains is understanding the key sources of ionizing radiation. To address this, our focus must shift to lower redshifts, where ionizing radiation can traverse IGM.

Several factors, such as low dust content, and higher ionizing escape fraction, contribute to why dwarf galaxies play a substantial role in ionizing the IGM. These parameters are influenced by the star formation history (SFH). Suggested by various observations and simulations, dwarf galaxies have a bursty SFH, characterized by significant variations in SFRs on short timescales ($\sim 5\text{-}10$ Myr), unlike their larger counterparts. Hence, to comprehend their ionizing capabilities, it is imperative to investigate their SFH.

We propose for archival funding to incorporate the JWST NIRISS grism data in Abell 2744 galaxy cluster, from GLASS ERS with deep existing HST/WFC3 UV data to investigate (1) how effective are dwarf galaxies in producing ionizing photons at $0.5 < z < 2.4$, (2) their bursty star formation

history. We will measure the UV luminosity from HST imaging and H-alpha

Luminosities from JWST spectroscopy for a sample of 326 lensed dwarf galaxies covering down to $\log(M^*)=7.5$ and $M_{uv}=-14$ at $0.5 < z < 2.4$. UV and H-alpha will determine star formation rates on very different timescales, allowing us to measure ξ_{ion} and to quantify the level of burstiness in these galaxies. We select Abell 2744 cluster, because it provides large lensing magnification, which enables the study of faintest galaxies that would otherwise be challenging to detect.

OBSERVING DESCRIPTION